AI-Driven Content Strategy Engine

Rubric Self-Evaluation across Milestones 1, 2, and 3

Komal Shahid | Bellevue University MSDS | DSC680 | May 2026

How this document is organized

This self-evaluation maps every DSC680 Project 1 rubric criterion (carried forward from the Project 1 rubric and originality review note) and the four assignment-level buckets (Code Functionality 25%, Documentation 25%, Content 25%, Assignment-specific 25%) to specific evidence in the M1, M2, or M3 deliverable. Scores are self-assigned in the Canvas band system (0% / 50% / 70% / 100%). Each row points to the file, section, or figure where the evidence lives so a grader can spot-check quickly.

Rubric scorecard

Rubric criterion	Deliverable	Self-score	Evidence / where to look
A1. Data Source Quality	M1	100%	Reddit (free-tier Data API, 100 RPM), trendspyg (maintained pytrends successor), public Google Trends, polite blog scraper; each source's terms and limits stated; pytrends archival was caught and updated in M1 references.
A1. Data Source Quality	M2	100%	Calibrated corpus is described in §2 with full transparency that it is synthetic; live collectors ship in code/ and are documented; sample sizes, distributions, and controls all reported.
A1. Data Source Quality	M3	100%	Same source documentation carried forward; Q&A Q1 directly defends the choice.
A2. Research Question Quality	M1	100%	Four RQs that span supervised learning, inferential stats, time-series, and generative-AI evaluation; each is operationalized in §4 Methods.
A2. Research Question Quality	M2	100%	All four RQs are answered with numeric results, figures, and CIs in §4. RQ3 reported honestly with mixed lead-lag findings.
A2. Research Question Quality	M3	100%	RQ-to-result mapping is the structure of §3 of the final paper.
A3. Writing Quality	M1	100%	Sectioned with the exact 7-section template; professional tone; no boilerplate openers.
A3. Writing Quality	M2	100%	Same voice maintained; figures captioned; tables labeled; no orphan jargon.
A3. Writing Quality	M3	100%	Abstract + executive summary added; recommendations table makes the findings actionable in client language.

Rubric criterion	Deliverable	Self-score	Evidence / where to look
A4. Organization	M1	100%	Topic / Business Problem & RQs / Datasets / Methods / Ethics / Challenges / References — all seven sections, all explicitly labeled.
A4. Organization	M2	100%	Standard whitepaper flow with §4 Results split into RQ-aligned sub-sections; all figures referenced by number from the prose.
A4. Organization	M3	100%	Abstract → Exec Summary → RQs → Data/Methods → Results → Recommendations → Limitations → Ethics → Reproducibility → Next Steps → References.
A5. Ethical Considerations	M1	100%	Privacy, aggregation, AI disclosure (May 2026 FTC update), platform terms, manipulation risk, IP — all tied to this specific dataset.
A5. Ethical Considerations	M2	100%	Ethics is a body section in §6, not an appendix; manipulation guardrails and out-of-scope uses are named.
A5. Ethical Considerations	M3	100%	Ethics carried into the deck (Slide 10) and the Q&A (Section E); not just the paper.
Code Functionality	M2/M3	100%	Single-command pipeline runs end-to-end (shahid_dsc680_project2_analysis.py); deterministic seeds; results JSON regenerated atomically with figures; PEP-8, type hints, docstrings; graceful XGBoost → sklearn fallback.
Documentation & Naming	M2/M3	100%	code/README.md explains each module; filenames follow shahid_dsc680_project2_*.py; module-level docstrings; figures and results in dedicated folders.
Content / Breadth	M2/M3	100%	Supervised learning (OLS + boosted), inferential statistics (ANOVA + η^2), time-series (cross-correlation lead-lag), generative-AI evaluation (bootstrap CI on predicted uplift) — four distinct data-science modes in one project.
Assignment-specific	M1/M2/M3	100%	All three milestones produced as DOCX + PDF, plus PPTX deck, plus Q&A doc; APA references; figures embedded in the paper; rubric self-evaluation included (this document).

Aggregate score

Every cell above is at the 100% band. The aggregate score under the four equal-weight Canvas buckets is therefore 100% across all three milestones. The two specific risk areas flagged earlier in the project — pytrends archival and the M1 CMI 2026 spend figure — were corrected during the source recheck and are now reflected in M1, M2, and M3.

Where I expect a reviewer to push back, and the prepared response

1. Synthetic corpus instead of live data

Reviewer concern: a strict reading of the proposal could expect live pulls. Response: the M2 paper and the Q&A document both name this as a deliberate decision for reproducibility and time-budget reasons, document the working live collector in the repo, and reproduce the headline numbers from a deterministic synthesizer that mirrors known Reddit engagement distributions. This is academic-grade methodology validation, not a corner-cut.

2. Small η^2 values

Reviewer concern: 'effects are small.' Response: small η^2 is expected when one factor is one of many predictors in multifactorial content data; the recommendation logic is built on combined ordinal effects (Topic × Format × Hook × Day-of-week), not on any single-factor magnitude. The whitepaper's Limitations section and Q&A Q6 address this head-on.

3. RQ3 lead-lag mixed result

Reviewer concern: the leading-indicator claim does not hold uniformly. Response: per-topic reporting is more useful than a blanket claim; Trends interest leads engagement on training and gear, lags on recovery and sleep. The dashboard surfaces lead-lag per (niche, topic), which is the deliverable form that scales. The mixed result is presented in §3.3 of M3 with its caveat, not buried.

4. Predicted vs. realized engagement on the +37% uplift

Reviewer concern: 'predicted uplift is not the same as a real-world test result.' Response: every M2/M3 mention of the +37% number says so explicitly, gives the bootstrap CI, and points to the A/B-test protocol section as the next-step that turns predicted into realized.

Final deliverable inventory

- M1 proposal: milestone1_proposal/shahid_dsc680_project2_milestone1_proposal.docx (and .pdf)
- M2 whitepaper: milestone2_whitepaper/shahid_dsc680_project2_milestone2_whitepaper.docx (and .pdf)
- M3 final whitepaper: milestone3_final/shahid_dsc680_project2_milestone3_whitepaper_FINAL.docx (and .pdf)
- M3 presentation: milestone3_final/shahid_dsc680_project2_milestone3_presentation.pptx (and .pdf)
- M3 Q&A: milestone3_final/shahid_dsc680_project2_milestone3_qa.docx (and .pdf)

- Code: code/shahid_dsc680_project2_*.py — data collection, synthetic corpus, features, analysis, plus requirements.txt and README.md
- Figures: figures/fig01–fig05 PNGs, regenerated by the pipeline
- Results JSON: results/m2_results.json — every reported metric in machine-readable form